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GR-CLASS: Comprehensive Technical & Functional SRS

1. System Blueprint

GR-CLASS is a **Maritime Classification & Certification ERP**. It manages the technical health, regulatory compliance, and official documentation for commercial vessels.

Core Tech Stack

- **Backend:** Node.js (v20+), Express.js (v5)
- **Database:** MySQL 8.0 (Sequelize ORM)
- **Caching:** Redis
- **Cloud:** AWS (S3 for documents, SES for Email)
- **Mobile Support:** Optimized API for offline-first surveyor apps.

2. User Roles & Permission Matrix

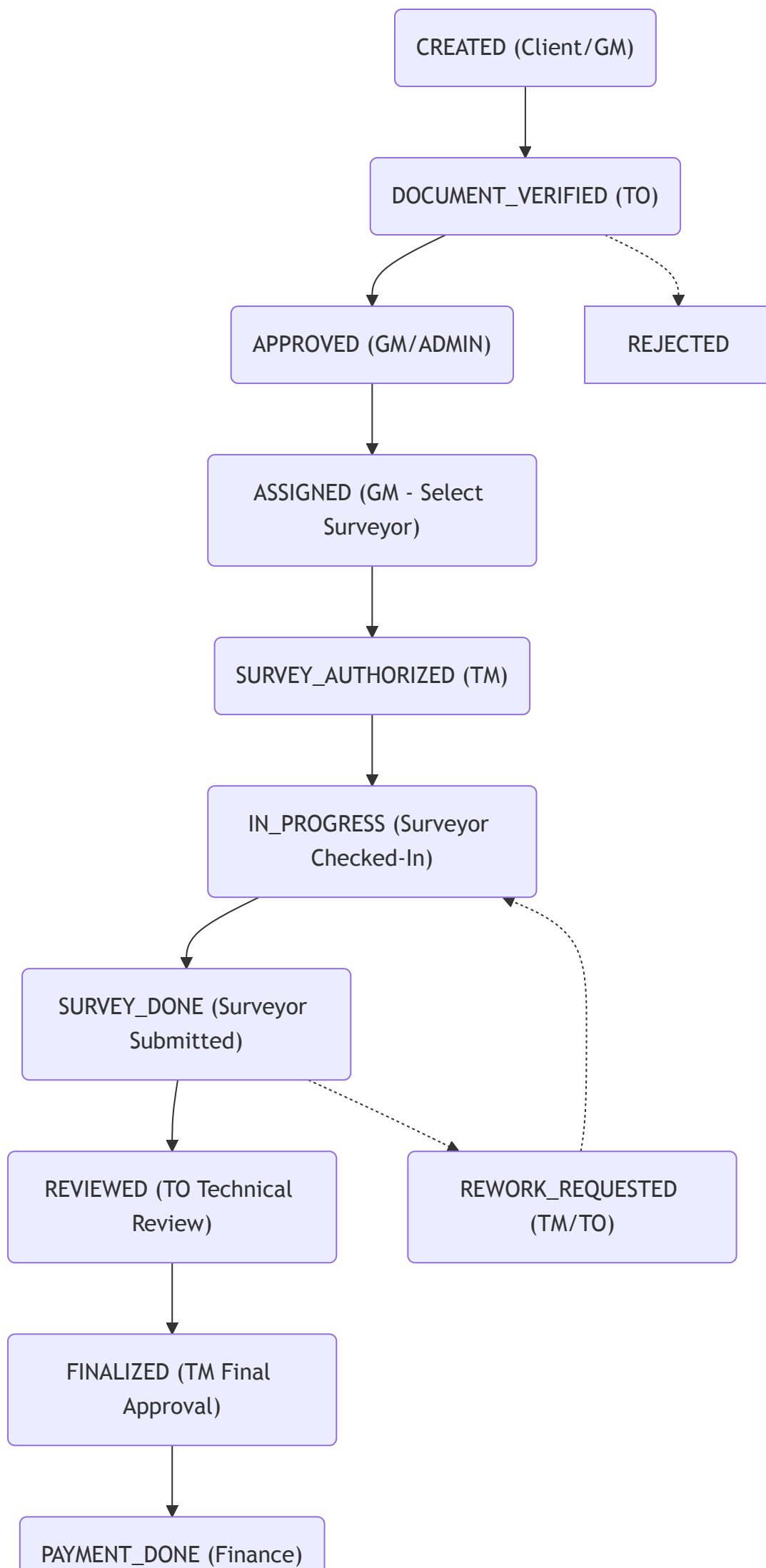
The system uses **Role-Based Access Control (RBAC)**. Each role has a specific "Data Scope".

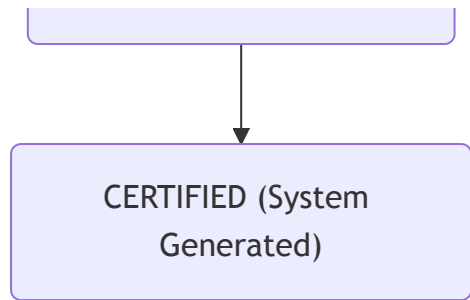
Role	Meaning	Responsibilities	Data Scope
ADMIN	Administrator	Total system control.	Global
GM	Gen. Manager	Business flows, Client/Surveyor onboarding, Job Approval.	Global
TM	Tech. Manager	Technical oversight, Final Survey Approval, Cert Issuance.	Global
T0	Tech. Officer	Document Verification, Technical review of checklists.	Global
TA	Tech. Assistant	Operational support, help with logs.	Global
SURVEYOR	Inspector	Performing vessel surveys, uploading reports.	Assigned Jobs Only
CLIENT	Ship Owner	Vessel management, requesting surveys, cert downloads.	Own Organization Only
FLAG_ADMIN	Port State	Regulatory oversight of vessels under their flag.	Flag-Specific

3. The "Engine": Workflow Cycles

3.1 Job Request Lifecycle

This is the primary state machine of the application.





3.2 Surveyor Field Workflow (Mobile App)

Surveyors follow a strict order to ensure data integrity:

1. **Authorize:** Admin authorizes the survey locally at a port.
2. **Check-in:** Surveyor hits `POST /surveys/start`. Start GPS and timestamp recorded.
3. **Checklist:** Surveyor fills technical questions (`PUT /checklists/jobs/:id`).
4. **Evidence:** Upload photos of engine, hull, safety gear (`POST /surveys/jobs/:id/proof`).
5. **Check-out:** Surveyor submits report (`POST /surveys`). Captures digital signature and finish GPS.

4. Module API Reference

4.1 Authentication (`/auth`)

- `POST /auth/login`: Identity verification.
- `POST /auth/refresh-token`: Renew short-lived access tokens.
- `POST /auth/logout`: Revoke tokens and clear cookies.
- `POST /auth/forgot-password`: Email-based reset.

4.2 Job Management (`/jobs`)

Method	Endpoint	Description	Roles
GET	/	List jobs with status/client filters.	All (Scoped)

Method	Endpoint	Description	Roles
POST	/	Request a new survey/cert.	CLIENT, GM, ADMIN
PUT	/:id/verify-documents	Mark docs as checked.	TO, GM, ADMIN
PUT	/:id/approve-request	Approve for surveyor assignment.	GM, ADMIN
PUT	/:id/assign	Link a Surveyor to the job.	GM, ADMIN
PUT	/:id/authorize-survey	Unlocks survey flow for surveyor.	TM, ADMIN
PUT	/:id/review	Mark technical review complete.	TO, ADMIN
PUT	/:id/finalize	Final signature before billing.	TM, ADMIN
PUT	/:id/priority	Update job urgency (URGENT/NORMAL).	GM, TM, ADMIN

4.3 Survey Operations (/surveys)

- **POST /surveys/start**: Record check-in coordinates.
- **POST /surveys/jobs/:id/proof**: Upload evidence photo (S3).
- **POST /surveys/jobs/:id/location**: Stream GPS pings during inspection.
- **POST /surveys/jobs/:id/sync**: Bulk upload offline-cached data.
- **PUT /surveys/jobs/:id/finalize**: TM command to lock the report.

4.4 Certification Engine (/certificates)

- **GET /certificates/types**: List available certs (e.g., Annual, Interim).
- **POST /certificates/**: Generate a draft from job data.
- **POST /certificates/:id/issue**: Converts draft to official Signed PDF.
- **GET /certificates/verify/:number**: Public endpoint for QR verification.
- **PUT /certificates/:id/renew**: Start renewal workflow (extends expiry).

4.5 Client & Vessel Management (/vessels, /clients)

- **GET /vessels**: List vessels (Filter by Flag, Age, Client).
- **POST /vessels**: Register new vessel info + IMO number.
- **GET /clients/:id**: View company details & primary contacts.
- **PUT /clients/:id/status**: Suspend/Activate organization.

4.6 Supporting Modules

- **Non-Conformities**: **GET /non-conformities** (List vessel deficiencies identified in survey).
 - **Checklists**: **GET /checklist-templates** (Admin config for survey questions).
 - **Dashboard**: **GET /dashboard** (Role-specific counters for Pending Tasks, Expiring Certs).
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5. Technical Constraints for Frontend Implementation

5.1 Success/Error Handling

The backend always responds with a standardized wrapper:

- **Status 200/201**: Success. `json.success = true`.
- **Status 400**: Validation Error (Check `json.message` for Joi details).
- **Status 401/403**: Auth/RBAC issues.
- **Status 404**: Item not found.

5.2 Document Uploads

The backend uses **Pre-signed URLs** for maximum performance.

1. Frontend requests **GET /api/v1/certificates/upload-url**.
2. Backend returns an S3 `uploadUrl` and a `fileKey`.

3. Frontend performs **PUT** to AWS S3 directly.
4. Frontend notifies Backend with **fileKey** in the final **POST**.

5.3 Offline Synchronization

Surveyors often work in areas with 0 connectivity. The Frontend should:

- Cache Checklist data locally (IndexedDB/Redux Persist).
 - Store GPS points with timestamps.
 - Use **POST /surveys/jobs/:id/sync** to replay the entire session once online.
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6. Role-Based Navigation Mapping (Checklist)

Client Panel

- ☐ View Dashboard (My Vessels, Active Jobs).
- ☐ Request New Job.
- ☐ Upload Vessel Supporting Docs.
- ☐ Download Issued Certificates.

Admin/Manager Panel (GM/TM)

- ☐ List all Client Requests (New).
- ☐ Assign Surveyors (Map integration recommended).
- ☐ Review Surveyor Submissions (Photo gallery view).
- ☐ Audit Logs & Report Builder.

Surveyor Panel (The "Field Unit")

- ☐ View Assigned Jobs Calendar.
 - ☐ Interactive Checklists (Toggle switch for Pass/Fail).
 - ☐ Camera Integration (Evidence).
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7. Complete Route Dictionary

(Exhaustive List)

Module	Method	Path	Required Role(s)
Auth	POST	/auth/login	Public
	POST	/auth/refresh-token	Public
	POST	/auth/logout	All
Vessels	GET	/vessels	ADMIN, GM, TM, TO, CLIENT
	POST	/vessels	ADMIN, GM, TM
	GET	/vessels/:id	ADMIN, GM, TM, TO, SURVEYOR, CLIENT
	PUT	/vessels/:id	ADMIN, GM, TM
Jobs	GET	/jobs	All (Scoped)
	POST	/jobs	CLIENT, ADMIN, GM
	PUT	/jobs/:id/verify-documents	ADMIN, TO, GM
	PUT	/jobs/:id/approve-request	ADMIN, GM
	PUT	/jobs/:id/assign	ADMIN, GM
	PUT	/jobs/:id/authorize-survey	ADMIN, TM
	PUT	/jobs/:id/review	ADMIN, TO
	PUT	/jobs/:id/finalize	ADMIN, GM, TM
Surveys	POST	/surveys/start	SURVEYOR
	POST	/surveys/jobs/:id/proof	SURVEYOR
	POST	/surveys/jobs/:id/location	SURVEYOR
	POST	/surveys/jobs/:id/sync	SURVEYOR
	POST	/surveys	SURVEYOR (Final Submission)
	PUT	/surveys/jobs/:id/finalize	ADMIN, TM

Module	Method	Path	Required Role(s)
Certificates	GET	/certificates/types	All (Scoped)
	POST	/certificates	ADMIN, GM, TM
	POST	/certificates/:id/issue	ADMIN, GM
	GET	/certificates/:id/download	All (Scoped)
	PUT	/certificates/:id/suspend	ADMIN, TM
	PUT	/certificates/:id/revoke	ADMIN, TM
Clients	GET	/clients	ADMIN, GM, TM, TO
	POST	/clients	ADMIN, GM, TM
	GET	/clients/:id	ADMIN, GM, TM, TO
Users	GET	/users/me	All
	GET	/users	ADMIN
	POST	/users	ADMIN
Payments	GET	/payments	ADMIN, GM, TM
	POST	/payments	ADMIN, CLIENT (Trigger)
Support	POST	/support/tickets	All
	GET	/support/tickets	ADMIN, GM (Staff view)
System	GET	/system/health	Public
	GET	/system/audit-logs	ADMIN
	GET	/system/metrics	ADMIN
Feedback	POST	/customer-feedback	CLIENT
	GET	/customer-feedback	ADMIN, GM
Search	GET	/search	All (Scoped)
Dashboard	GET	/dashboard	All (Role-specific)